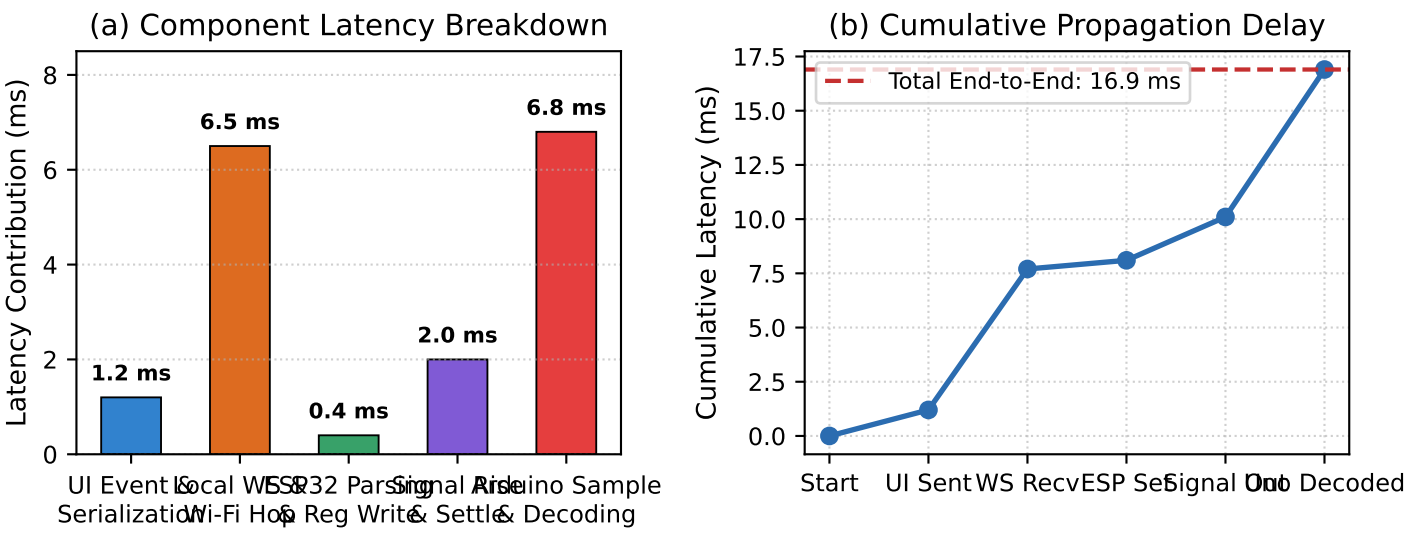
(a) Component Latency Breakdown
Latency Contribution (ms)
0
2
4
6
8
1.2 ms
6.5 ms
0.4 ms
2.0 ms
6.8 ms
UI Event Serialization
Local Wi-Fi Hop
ESP32 Parsing & Reg Write
Signal Rise & Settle
Arduino Sample & Decoding
(b) Cumulative Propagation Delay
Cumulative Latency (ms)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Total End-to-End: 16.9 ms
Start
UI Sent
WS Recv
ESP Set
Signal Out
Decoded